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Unit 5 : Session 5 : SLO 2 | ARM7 Processor

I. Multiple Choice

1. b) It is a 32-bit RISC processor.
2. c) 3-stage pipeline (Fetch, Decode, Execute).
3. c) Both ARM (32-bit) and Thumb (16-bit) instruction sets.
4. b) Store the status of the processor, including flags and mode.
5. b) The PC (Program Counter) is updated to a new address.

II. True or False

1. TRUE.
2. TRUE.
3. FALSE — ARM7 uses a 3-stage pipeline (ARM9 moved to 5 stages).
4. TRUE.
5. FALSE — a branch writes the target address into the PC.

III. Scenario-Based Activity

1. Thumb benefit: 16-bit Thumb instructions roughly halve code size, so the firmware fits smaller/cheaper flash and makes better use of memory bandwidth — ideal for a memory-constrained embedded sensor device.
2. Instruction execution: handled by the datapath under the Control Unit — the CU decodes each instruction and drives the ALU/registers; the ALU performs the actual arithmetic/logical operations, and the CPSR records the resulting flags.
3. Data hazards: in the 3-stage pipeline an instruction may read a register before a previous instruction has written its result (RAW hazard), producing wrong values or stalls. Mitigation: insert NOPs/reorder instructions (software scheduling), since ARM7 has no forwarding; the compiler's instruction scheduler usually does this.